

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460816.8875 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.023
Transit depth (Rp/Rs)^2	3.08 +/- 0.79 [%]
Orbital Inclination (inc)	81.98 +/- 1.3
Airmass coefficient 1 (a1)	1.026 +/- 0.019
Airmass coefficient 2 (a2)	-0.02 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.83 %
Best Comparison Star	#2 - [207, 263]
Optimal Aperture	2.96
Optimal Annulus	6.55
Transit Duration (day)	0.0713 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.023 vs 0.139 (deviation 1.18σ)
Duration fit vs published	0.0713 d vs 0.0687 d (deviation 0.2σ)
Mid-time O–C	1.9 min
Depth SNR	5.7
Residual scatter	1.83 %

Light curve

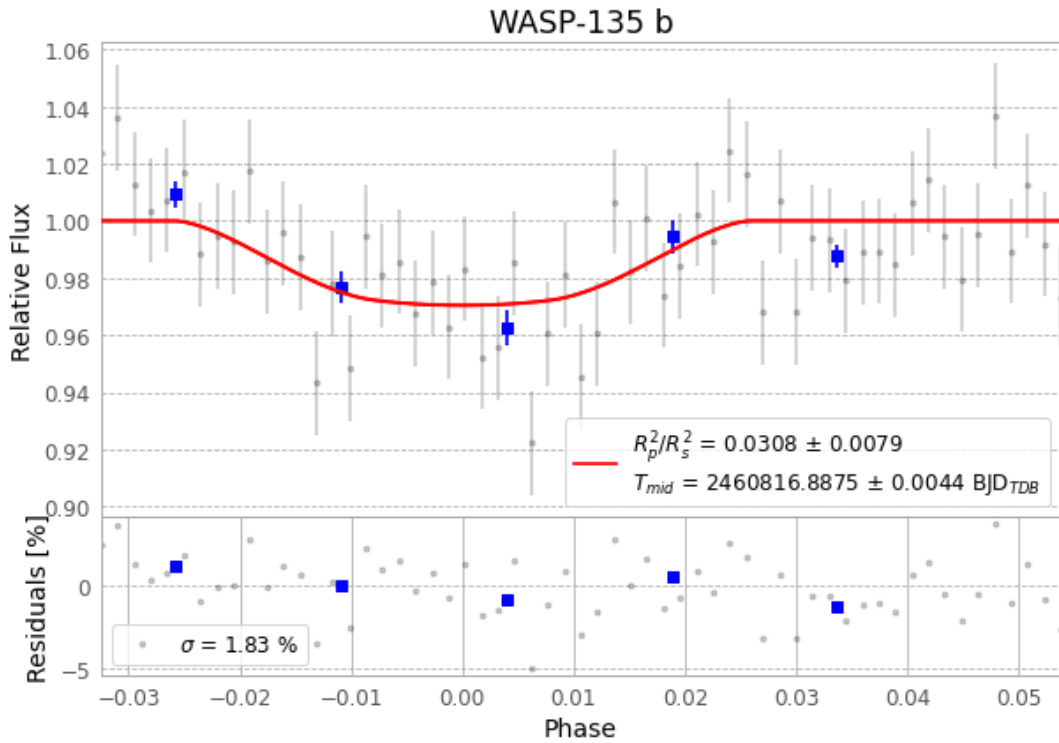


Figure 1 — FinalLightCurve_WASP-135 b_2025-05-21.png (SHA-256 7eca6465f7f2d773...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 06ef601c704d53dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

59 FITS frames (39 MB), WASP-135250521080615.FITS → WASP-135250521110016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250521.log (3b0f64ce33c7...), FinalLightCurve_WASP-135 b_2025-05-21.png (7eca6465f7f2...), FinalLightCurve_WASP-135 b_2025-05-21.csv (ed7c03ee0524...)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:24Z.